# Chapter 9 — AI Agents

*Last reviewed: September 2026. Frameworks and products change every few months; the ideas (loop, tools, bounds, memory, evaluation, control) change slowly.*

Until now the model **answers**. An **AI agent**'s output also **does things** — looks up an order, runs code, sends an email. A wrong answer misleads a reader; a wrong action changes the world, sometimes irreversibly.

**What you'll learn:** what an agent is and when to use one; the ReAct loop and its budgets; tool calling and tool design; guardrails and human approval; planning; memory; multi-agent patterns; MCP (Model Context Protocol, a standard for plugging tools and data into agents); evaluation; rollout, cost and failure modes; a full design walk-through.

Code is in `code/`; each script runs offline in about a second with `python code/<file>.py`. The "LLM" there is a **deterministic** (same input → same output) scripted stand-in so you can see every step; real-API versions are marked "(needs an API key — not run)" (an **API key** is the secret string identifying you to a paid service). Interview answers are collected at the end.

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## 9.1 What an agent is

### What it is

An **AI agent** pursues a goal by repeatedly deciding what to do next, doing it through **tools**, observing the result, and deciding again; the decider is usually a large language model (LLM).

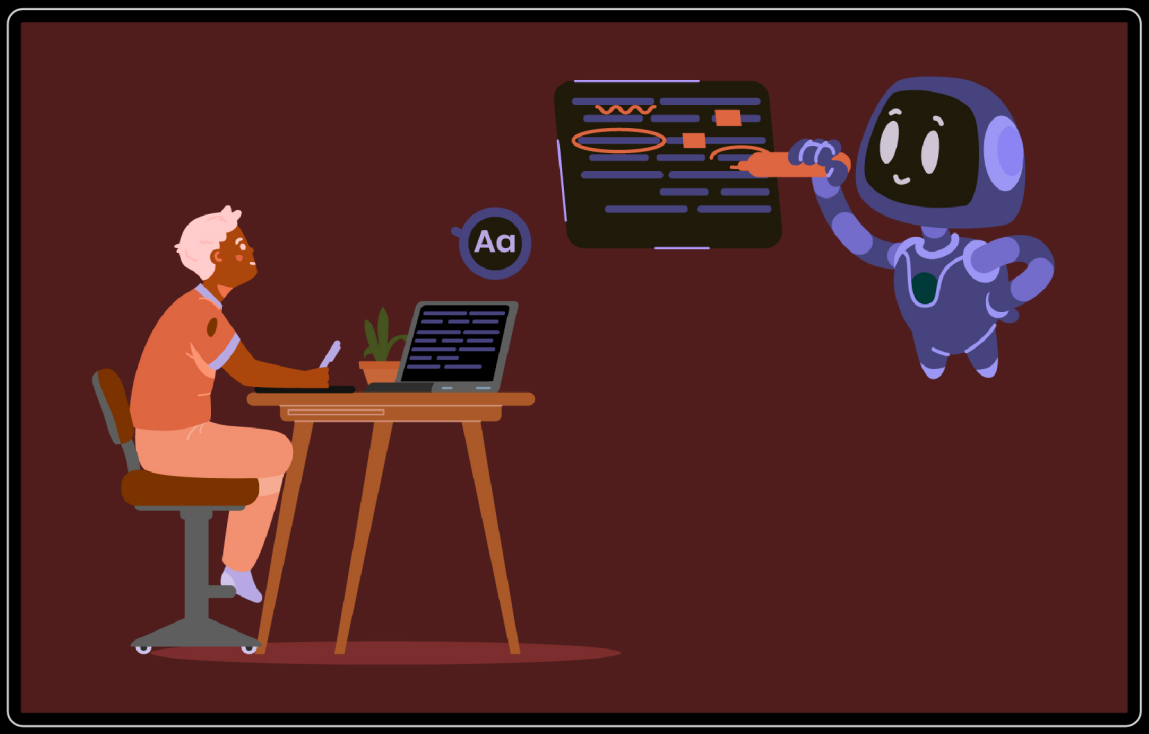



*Figure — agentic AI as a collaborator rather than a single-shot tool (course diagram).*

| Part | What it is |
|---|---|
| **Model (LLM)** | Reads the situation, proposes the next step (the employee's judgment) |
| **Tools** | Functions it may request: search, calculator, database lookup, email, run code |
| **Memory** | The context window (all text sent in one call) plus what it can look up (notes, past cases, preferences) |
| **Loop** | Code that calls the model, runs the requested tool, feeds back the result, repeats until done or stopped |

**Key terms.** The model never runs a **tool**: it emits a structured request ("call `lookup_order` with `order_id="4711"`") and the **runtime** (or **harness** — loop, tool execution, validation, permissions, logging) decides whether to run it. The result fed back is an **observation**; the ordered record of one task is its **trajectory**; what the agent may do is its **action space**.

### Agent vs chatbot vs workflow

A **chatbot** is *reactive*: asked about London's weather, it answers; an agent might also add "pack an umbrella" to your to-do list — it serves the goal behind the question (generative AI is the brain; the agent adds hands). In a **workflow** an engineer fixed the steps at design time (*transcribe → summarise → extract action items*); the LLM fills each step but never chooses the next.

> **Who chooses the next step?** Fixed in code at design time → workflow. Picked by the model at run time from what it found → agent.

### The autonomy spectrum

**Autonomy** — how much the system may finish before a person looks — is a dial: **fixed workflow** → **router** (model picks branch A or B) → **agent whose writes a person approves** → **bounded autonomy** (acts inside hard limits, e.g. refunds under $30; people audit samples). Maturity levels: 9.12.

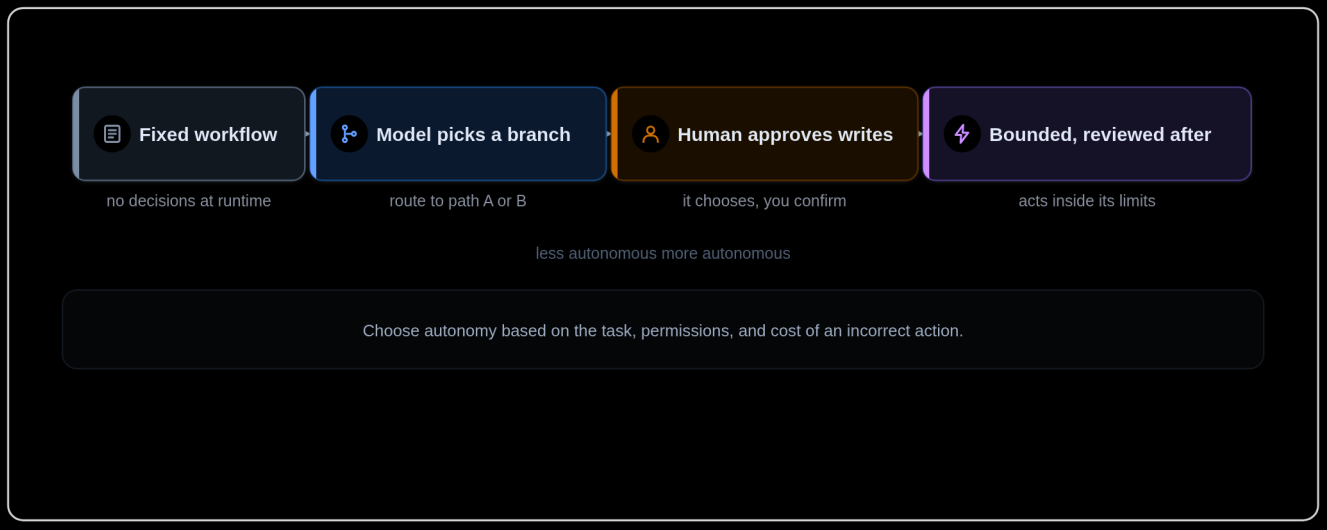



*Figure — say where your design sits (course diagram).*

Agents handle **multi-step tasks hard to specify in advance** (diagnosing a support problem, fixing a failing test, research) at the price of cost, **latency** (request time, end to end — System Design Fundamentals) and new failure modes (9.10).

### How it works: "agent or workflow?"

1. **List the steps.** If you can draw the flowchart with every branch, build a workflow: cheaper, faster, easier to test and budget.
2. **Find the open-ended part** (diagnosis, debugging, research) where the next step depends on what was just learned; only there does an agent earn its cost.

3. **Build a hybrid**: fixed code for intake, validation, post-processing around a small action space.
4. **Justify every extra tool, agent or loop** with measured quality gains.

Anthropic's *Building effective agents* (December 2024) agrees: one call, then a workflow, an autonomous agent only when those fall short.

### Trade-offs

| | Single LLM call | Workflow | Agent |
|---|---|---|---|
| Who decides steps | — | Engineer, design time | Model, run time |
| Cost and latency | Lowest | Low, predictable | Higher, variable (seconds to minutes) |
| Testability | Easy | Easy (known paths) | Hard (many paths) |
| Handles surprises | No | Only foreseen ones | Yes, within its tools |

**Avoid** for stable known steps, irreversible costly mistakes with no approval step, sub-second latency budgets, or when one well-prompted call works. **Mistakes:** calling every LLM feature an "agent"; an agent for the whole product when only 30–40% of cases are open-ended; every tool "just in case" (bigger action space, more ways to fail).

### Self-check

1. Workflow vs agent? → Who picks the next step: code at design time, or the model at run time.
2. Does the model execute tools? → No; it emits a request; the runtime validates and runs it.

### TL;DR

Agent = LLM + tools + memory + loop; the model proposes, code decides and acts. Use one only for the open-ended part.

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## 9.2 The agent loop and ReAct

### What it is

Every agent runs **reason → act → observe → repeat**.

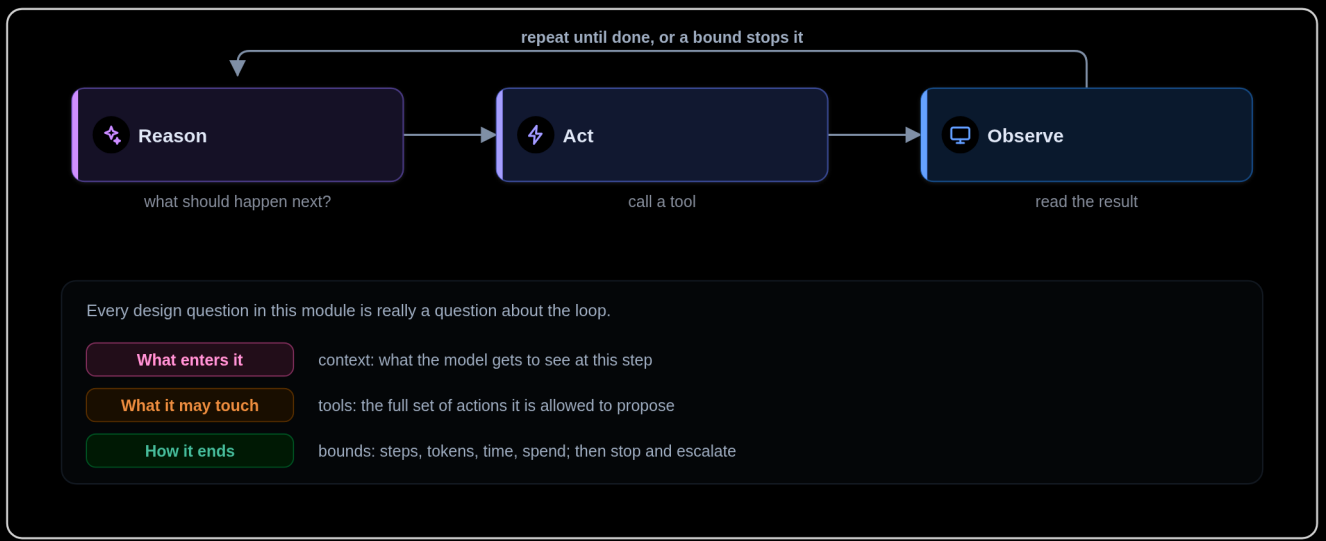



*Figure — the loop and its three design questions: what enters it, what it may touch, how it ends (course diagram).*

**ReAct** ("Reason + Act", Yao et al., 2022) prompts the model to write:

```
Thought: what I know and what I should do next
Action: the tool to call
Action Input: {"arg": "value"}
```

The runtime appends `Observation: <result>` and repeats until `Final Answer: ...`. Interleaving lets each observation correct the plan (reasoning alone invites made-up facts; acting alone, aimless calls). Modern APIs use structured tool calls instead of text (9.3); the loop is the same.

### Why it matters: the loop has no built-in brakes

Nothing guarantees it will **end**, **make progress** or **stay cheap**:

1. **Errors compound.** At 95% independent per-step success, 10 steps succeed $0.95^{10} \approx$ **0.60** of the time, 20 steps 0.36. Errors aren't fully independent and retries help, so measure end to end — but an early mistake becomes every later step's premise.
2. **Cost grows faster than steps.** Each call resends the growing history; total input **tokens** (billed word pieces, ~¾ word each) over n steps $\approx n(n+1)/2 \propto n^2$. **Prompt caching** (cheaper reuse of a repeated prefix) and compaction (9.6) reduce this.
3. **Agents fail quietly** — a stuck agent repeats calls and burns tokens rather than crashing.

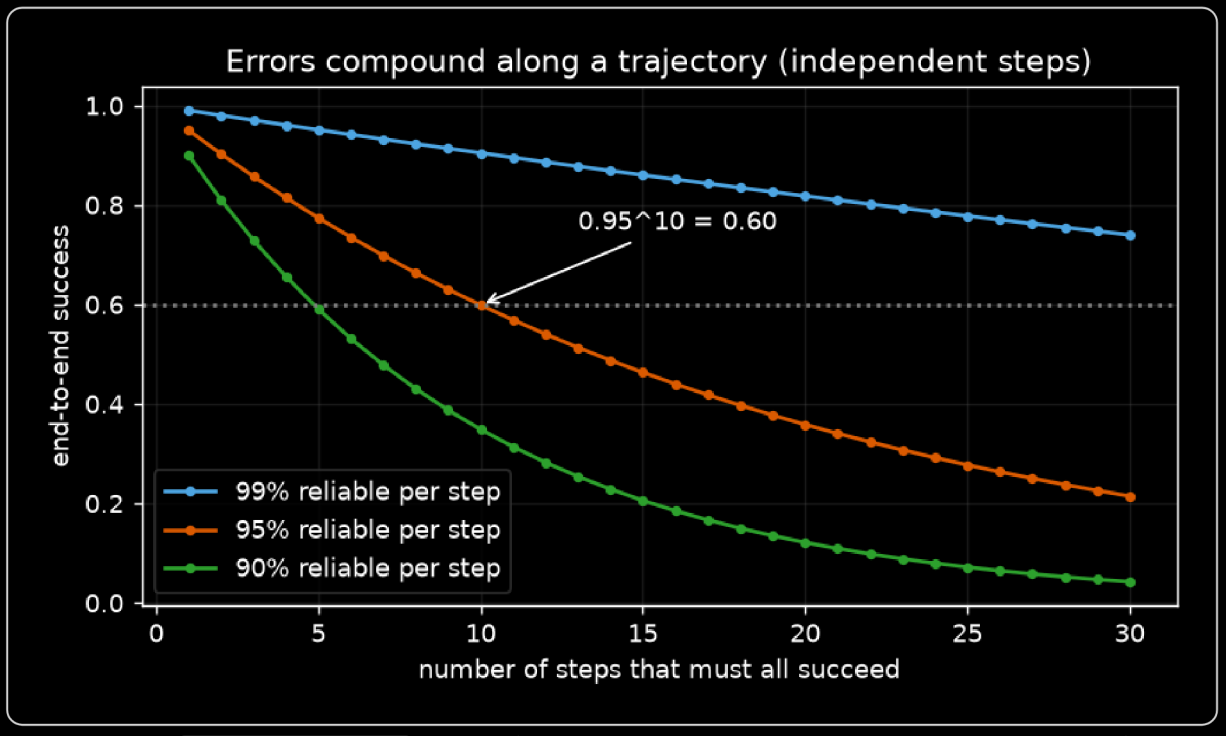



*Figure — from `code/agent_eval.py`: $0.95^{10} \approx 0.60$; even 99%-reliable steps give $0.99^{30} \approx 0.74$.*

So **every loop ships with budgets** (steps, tokens or dollars, wall-clock time, caps on consequential actions) and an exhaustion behaviour: **halt, save state, summarise progress, hand to a person**.

### How it works: a complete ReAct agent in pure Python

`code/react_agent.py` has a calculator that walks Python's **syntax tree** (never `eval()`, which would run any code the model wrote) and a TF-IDF-style search over five fictional handbook pages. `run_tool()` turns unknown tools, bad arguments and exceptions into `ERROR ...` observations.

```python
def react(question: str, llm, max_steps: int = 6) -> AgentResult:
    # ... build the prompt: instructions + tool descriptions + question
    for _ in range(max_steps):                      # bound 1: step budget
        reply = llm.complete(transcript)
        try:
            thought, action, args, final = parse(reply)
        except (ValueError, json.JSONDecodeError) as e:
            transcript += f"{reply}\nObservation: ERROR could not parse reply ({e})\n"
            continue
        if final is not None:
            return AgentResult(final, "final_answer", steps)
        key = f"{action}:{json.dumps(args, sort_keys=True)}"
        if key in seen_actions:                     # bound 2: no-progress detector
            return AgentResult("I could not find this; escalating to a human.",
                               "repeated_action", steps)
        seen_actions.add(key)
        observation = run_tool(action, args)
        transcript += f"{reply}\nObservation: {observation}\n"
    return AgentResult("Step budget used up; escalating to a human.", "max_steps", steps)
```

The `ScriptedLLM` stand-in receives exactly the prompt a real model would and reproduces two classic failures on purpose. Output (excerpt):

```
Q: How much do 3 Pro plan seats cost per year if we pay annually?
  [1] Thought: I need the Pro plan price first.   (prompt ~120 tokens)
      Action: search_docs {"query": "Pro plan price"}
  [2] Thought: Pro is $20, annual discount 15%. Compute 3 seats x 12 months.   (prompt ~207 tokens)
      Action: calculator {"expression": "3 * 20 * 12 * (1 - 15/100)"}
      Observation: 612
  [3] Thought: The calculator returned 612.   (prompt ~248 tokens)
  => 3 Pro seats billed annually cost $612 per year.   [stopped: final_answer]
  total input tokens sent across steps: ~575

Q: How long does delivery to Canada take?
      Observation: ERROR unknown tool 'lookup_shipping'. Available tools: ['calculator', 'search_docs']
  => Delivery to Canada takes 5-8 business days.   [stopped: final_answer]

Q: What is the CEO's favourite colour?
  => I could not find this; escalating to a human.   [stopped: repeated_action]
```

- **Q1**: maths goes to the calculator (720 × 0.85 = 612 ✓); prompts grow 120 → 207 → 248 because the transcript is resent; the sum (~575) is the input cost.
- **Q2**: a **hallucinated tool** crashed nothing; the error listed the real tools and the agent recovered.
- **Q3**: the answer isn't in the documents; the repeat detector stops after two identical searches — honest failure beats confident invention.

## Trade-offs

Text ReAct works with any model and shows its reasoning, but **parsing** free text is fragile; **native tool calling** (9.3) returns schema-checked **JSON** and is the production default.

## Common mistakes

No step limit; a tool exception crashing the loop instead of becoming an observation; trusting "I'm done" without checking the goal (tests pass, record exists); letting the model do arithmetic, date maths or lookups from memory.

## Self-check

1. Why does input grow faster than steps? → Each call resends the history; $1 + \dots + n \approx n^2/2$.
2. What should an exhausted budget trigger? → Halt, save state, summarise, hand to a person.

## TL;DR

Reason → act → observe, inside budgets; failures become observations, repeats are caught, completion is verified.

## 9.3 Tool (function) calling

### What it is

**Function calling** ("tool use") lets a model request a tool in machine-checkable form. Each tool has a name, a description and a **JSON Schema** for its inputs (fields, types, required, allowed values `enum`, ranges `minimum`, whether extra fields are allowed `additionalProperties`). The response then contains e.g. `{"name": "get_order", "input": {"order_id": "A-100"}}` and generation stops; your code runs the tool and returns a **tool result**.

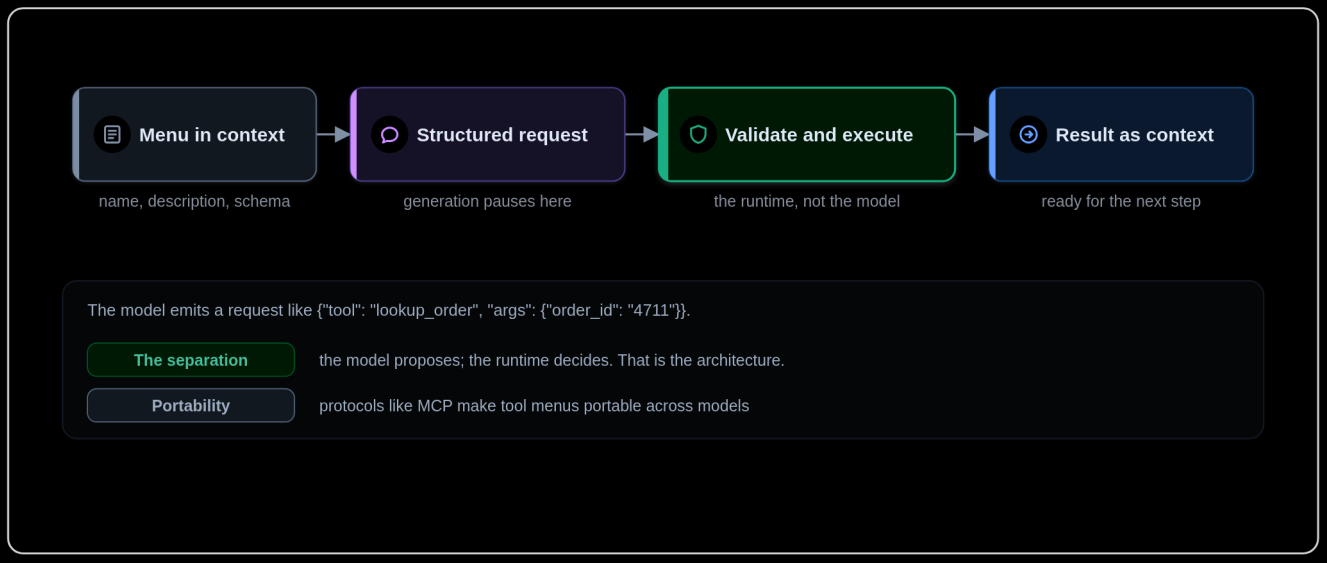



*Figure — menu in context → structured request → runtime validates and executes → result back as context (course diagram).*

### Why it matters: the model proposes, the runtime decides

This split is every agent's security architecture: the model can only **ask**; trusted code decides whether the request is valid, allowed and safe, so validation, permissions and audit logs live where no prompt can rewrite them.

### How it works

1. **Call the model** with the conversation and tool list.
2. If it stopped for tools, read the tool-call blocks — possibly **several**, to run in parallel.
3. **Validate and execute** each (schema, permissions, policy — 9.4).
4. **Return all results** in the next message, each linked by id, failures marked as errors; **loop** until a final answer or a budget stops you.

| Concept | Anthropic Messages API | OpenAI Chat Completions |
|---|---|---|
| Tool definition | `{"name", "description", "input_schema"}` | `{"type": "function", "function": {"name", "description", "parameters"}}` |
| Model asks for a tool | block `type: "tool_use"` with `id`, `name`, `input` (a dict) | `message.tool_calls[i]` with `id`, `function.name`, `function.arguments` (a JSON *string*) |
| Signal to run tools | `stop_reason == "tool_use"` | `finish_reason == "tool_calls"` |
| Send the result back | user message with `type: "tool_result"`, `tool_use_id`, `content`, optional `is_error` | message `role: "tool"` with `tool_call_id`, `content` |
| Guarantee schema-valid arguments | `strict: true` on the tool | `strict: true` on the function |

### Code: the real API version (needs an API key — not run)

The same loop with native tool calling in the Anthropic Python **SDK** (official client library; set-up in Chapter 6):

```
# (needs an API key — not run)
import anthropic

client = anthropic.Anthropic()          # reads ANTHROPIC_API_KEY
tools = [
    # {"name": "search_docs", ...} is defined the same way
    {"name": "calculator",
     "description": "Evaluate an arithmetic expression exactly. Use for all maths.",
     "input_schema": {"type": "object",
                      "properties": {"expression": {"type": "string"}},
                      "required": ["expression"], "additionalProperties": False}},
]
messages = [{"role": "user",
             "content": "How much do 3 Pro plan seats cost per year if we pay annually?"}]

for step in range(8):                                  # bounded loop
    response = client.messages.create(
        model="claude-opus-5-5",                       # model ids change; check the models page
        max_tokens=16000, tools=tools, messages=messages)
    if response.stop_reason != "tool_use":             # final answer (or other stop)
        break
    messages.append({"role": "assistant", "content": response.content})
    results = []
    for block in response.content:
        if block.type == "tool_use":
            out = run_tool(block.name, block.input)      # same validation as before
            results.append({"type": "tool_result", "tool_use_id": block.id,
                            "content": out, "is_error": out.startswith("ERROR")})
    messages.append({"role": "user", "content": results})   # all results in ONE message

print("".join(b.text for b in response.content if b.type == "text"))
```

Append the full `response.content` (not just its text); return all of one turn's results in one message. SDK "tool runner" helpers write this loop; you still own validation, permissions and budgets.

### Tool design: an API for an easily confused client

The consumer rereads the docs every call, pays for them in tokens and confuses them easily:

1. **Few, coarse, task-shaped tools.** `search_orders`, `get_order_details`, `issue_refund(order_id, amount, reason)` beat twenty low-level tools — and are **safer**: `issue_refund` enforces a cap and eligibility in code; a general `run_sql` can't be limited to legitimate refunds.
2. **Descriptions are prompts.** Say when to use a tool *and when not to* ("not for other customers' orders; use `escalate`"). Measure tool choice on a **golden set** (fixed cases with known answers) of "scenario →

expected call"; re-run when descriptions change.

3. **Errors are information.** A code (`INVALID_ORDER_ID`), what a valid value looks like, whether retrying makes sense. Vague errors cause retry loops; **stack traces** leak internals. Add **timeouts** and **rate limits**.
4. **Writes must be idempotent** (twice = once; System Design Fundamentals). Derive an **idempotency key** from the operation (case + order + amount), reuse it on retries; the service returns the stored result for duplicates.

**Hallucinated arguments** (invented tools or fields, `100` instead of `"A-100"`, values nobody gave) — defences, cheapest first: schema validation rejecting extra fields (strict mode constrains generation to the schema); world checks, since schema-valid is not *true* (does this order exist and belong to this user?); precise errors for self-correction; ids the runtime already knows rather than ids the model types.

### Common mistakes

A general-purpose tool (shell, any-table SQL, any-URL HTTP) where a narrow one would do; business rules only in the prompt; a fresh random idempotency key per retry (protects nothing); returning results for only some parallel calls, or across several messages.

### Self-check

1. How is a result linked to its call (Anthropic)? → `tool_use_id` = the `tool_use` block's `id`.
2. What makes a retried `issue_refund` safe? → A key derived from the logical operation, deduplicated by the service.

### TL;DR

Name + description + JSON Schema; the model requests, the runtime decides. Few, task-shaped, well-described tools; idempotent writes.

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## 9.4 Guardrails, permissions and human-in-the-loop

### What it is

**Guardrails** keep behaviour in bounds: input filters, output filters and — for agents — an **action gate** every proposed tool call passes before executing. **Human-in-the-loop (HITL)**: a person approves (or later reviews) certain actions. Guardrails and prompt injection in depth: §12.4, §12.6; here, what changes when the model can act.

### The problem it solves

A model error can now move money or delete data, and **prompt injection** — text crafted to override instructions, typed by the user (*direct*) or hidden in an email, page or document it reads (*indirect*) — can steer it. No prompt reliably prevents this, so the defence sits **outside the model**.

### How it works: three permission principles

1. **Least privilege, per task. Credentials** (keys, tokens proving who calls) only for this task's tools: *this customer's* orders and capped refunds, not arbitrary tables.
2. **The agent acts as someone, never beyond them.** Tools run under an authorised identity (the signed-in user, or a narrow **service role** — non-human account with fixed permissions) with short-lived per-task credentials; authority never comes from model output. Working for Alice, Bob's orders are unreachable whatever text it reads.
3. **Tier tools by consequence**, each tier with a control:

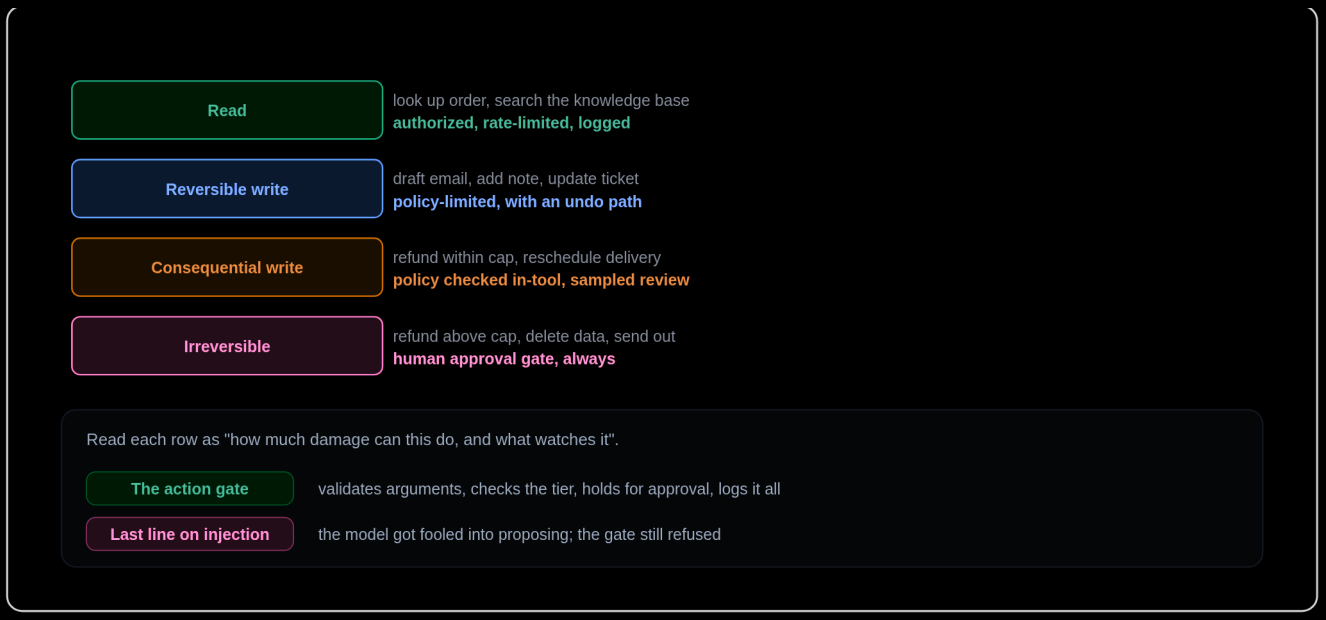



*Figure — four tiers, each with a matching control (course diagram).*

| Tier | Examples | Control |
|---|---|---|
| Read | Look up an order, search the knowledge base | Authorised reads only; rate-limited; logged |
| Reversible write | Case note, draft email, ticket status | Within policy; logged; undo path |
| Consequential write | Refund within a cap, reschedule a delivery | Policy checked inside the tool; sampled review |
| Irreversible / high-stakes | Refund above the cap, delete data, send externally | **Human approval, always** |

### The action gate, step by step

1. **Validate** against the schema.
2. **Authorise**: may this principal use this tool on this object now?
3. **Business rules**: caps, eligibility, per-conversation action limits.
4. **Classify risk**; if approval is needed, show the reviewer the *exact* tool and arguments, bind the approval to them, re-check just before executing.
5. **Execute** idempotently, then **log** decision, reason and approver (sensitive data **redacted**).

A fooled model can still *propose* a harmful action; the gate refuses it.

### Code: an action gate in action

`code/guarded_tools.py`: three tiered tools; a scripted model proposes ten calls, several deliberately bad. The core:

```python
def action_gate(call: dict, s: Session) -> dict:
    """Validate -> authorise -> apply policy -> maybe ask a human -> execute -> log."""
    tool, args = call["name"], call["input"]
    problems = validate(args, TOOLS[tool]["input_schema"])
    if problems:  # readable, structured error the model can fix
        return done({"ok": False, "error": "INVALID_ARGUMENTS", "details": problems,
                     "retryable": True}, "rejected")
    # ... missing OR not this customer's order -> ORDER_NOT_FOUND (leak nothing)
    if TOOLS[tool]["tier"] == "consequential":
        key = refund_key(s.case_id, args)
        if key in REFUNDS:  # idempotent replay: same result, no new money moves
            return done({"ok": True, "duplicate": True, "refund": REFUNDS[key]}, "deduplicated")
        # ... per-session action cap; amount <= what is still refundable
        if args["amount"] > REFUND_CAP and not human_approves(tool, args):
            return done({"ok": False, "error": "APPROVAL_DECLINED"}, "declined_by_human")
    return done(execute(tool, args, s), "executed")
```

| Step | Failure mode | Defence that fired |
|---|---|---|
| 1, 6 | Wrong type, invented field, invalid enum | Schema validation with a precise error |
| 3 | Injected request to refund another customer's $900 order | Authority check ("not found" reveals nothing about Bob) |
| 5 | Network retry of the same refund | Idempotency key → deduplicated |
| 7 | Refund larger than what is left on the order | Business rule in the tool path |
| 8–9 | Refunds above the $30 cap | Human approval: one approved, one declined |
| end | Too many steps | Step budget → stop and escalate |

It ends `Audit summary: {'declined_by_human': 1, 'deduplicated': 1, 'denied': 2, 'executed': 4, 'rejected': 2}`; only $72 moved ($12 + $60). The model was wrong repeatedly; the system never was.

### Other guardrails

**Sandboxes** for code-running tools (isolated **containers**, network **allow-lists**); a **kill switch** per tool; **alarms** on action rates (a spike may be attack, bug or genuine surge — investigate first). Measure approvers too: one who approves everything unread is not a control. Against indirect injection, treat fetched content as data; for high-risk agents, separate the part that reads untrusted content from the part that acts, or require approval for actions that follow such reading. Approval costs reviewer time and latency, so reads and undoable writes rarely need it (log, sample, undo) — keep it for irreversible or high-stakes actions.

### Common mistakes

Safety in the system prompt ("you must never …"); checking a tool is allow-listed, not that *this user* may do *this action* to *this object*; approval screens that hide the consequence ("Approve action #4?" vs "Refund $60 to order A-102"), or letting the call change between approval and execution.

### Self-check

1. Why "not found" for another user's order? → To avoid confirming it exists.
2. The four tiers? → Read, reversible write, consequential write, irreversible/high-stakes.

### TL;DR

The model can be fooled; the runtime cannot be argued with. Least privilege, tiers, an action gate, targeted approval.

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## 9.5 Planning: plan-and-execute and reflection

### What it is

**Planning** decides a sequence of steps before (or while) acting: **set a clear goal** ("book a room for 6 on Friday afternoon"), **reason** about what is known, **make a plan** of concrete steps, **adapt** when a step fails. Pure ReAct calls the LLM before every step — flexible, but it wanders and costs more; a plan gives structure, lets cheap code execute steps and makes progress checkable.

### How it works: the main patterns

| Pattern | Idea | Trade-off |
|---|---|---|
| **ReAct** (9.2) | One step at a time from the latest observation | Adapts instantly; many LLM calls, can drift |
| **Plan-and-execute** | Planner writes the plan, executor runs it, re-plan on failure | Fewer calls, readable plan; plan may be wrong from the start |
| **ReWOO** (Xu et al., 2023) | Plan all tool calls with placeholders, run them, one "solver" call answers | Very few calls; cannot react mid-way |
| **Reflection** | Critique the result and retry | Catches own errors; may agree with itself |
| **Reflexion** (Shinn et al., 2023) | Store a short verbal **lesson** per failure, reuse it next attempt | Learns without **retraining** weights; needs a reliable success signal |
| **Tree search** (Tree of Thoughts, Yao et al., 2023) | Explore candidate steps, back up from bad ones | Good for puzzles; costly |

Rule: **verify the world, not the agent's claim**. "I booked the room" is a claim; a booking record with enough capacity is a fact.

### Code: plan, execute, verify, reflect, re-plan

`code/plan_execute_reflect.py` books a room for 6: the planner first makes the naive choice; a code **verifier** checks real bookings; **reflection** turns each failure into a one-line lesson fed to the next plan; a **compensating action** (a step reversing an earlier step's effect) undoes a wrong booking. Output (excerpt):

```
Attempt 1: plan = [('book', {'name': 'Aspen', 'slot': 'fri-pm', 'people': 6})]
   verifier: FAIL (Aspen fits only 4 people)
   compensating action: cancelled the wrong booking
   reflection -> lesson stored: 'Filter rooms by capacity >= party size before booking.'
Attempt 2: plan = [('book', {'name': 'Birch', 'slot': 'fri-pm', 'people': 6})]
   run book{'name': 'Birch', 'slot': 'fri-pm', 'people': 6} -> FAILED: Birch already booked for fri-pm
   reflection -> lesson stored: 'Check availability before booking; do not assume a room is free.'
Attempt 3: plan = [('is_free', {'name': 'Cedar', 'slot': 'fri-pm'}), ('book', {'name': 'Cedar', 'slot': 'fri-pm',
'people': 6})]
   verifier: PASS (verified: Cedar holds 10 >= 6)
```

Attempt 1 "succeeded" as far as the tool knew (`BOOKED`) but missed the goal; only the independent verifier caught it. Attempt 3, with two lessons, checks first.

### When not to use it

Heavy planning is wasted on one- or two-call tasks. Reflection needs a real success signal (tests, checks, a judge validated against people); without one the model may just rephrase its mistake. **Mistakes:** the agent grading its own homework; unbounded "try until it works" loops; vague unverifiable steps ("research the topic"); not undoing a failed attempt's side effects.

### Self-check

1. What does the verifier check? → Actual world state against the goal.

2. Why can plan-and-execute be cheaper than ReAct? → One planning call instead of one per step.

### TL;DR

Goal → plan → act → verify against the world → reflect → re-plan, within an attempt budget.

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## 9.6 Memory: short-term, long-term and durable state

### What it is

**Short-term (working) memory** is the **context window**. **Long-term memory** is stored outside the model and brought back when relevant — preferences, facts, past outcomes. History piles up and costs tokens, and **more context is not better context** — models use the middle of long prompts less reliably ("lost in the middle", Liu et al., 2023); no window survives a crash, resumes tomorrow or remembers the user next week.

### How it works: four tiers plus durable state

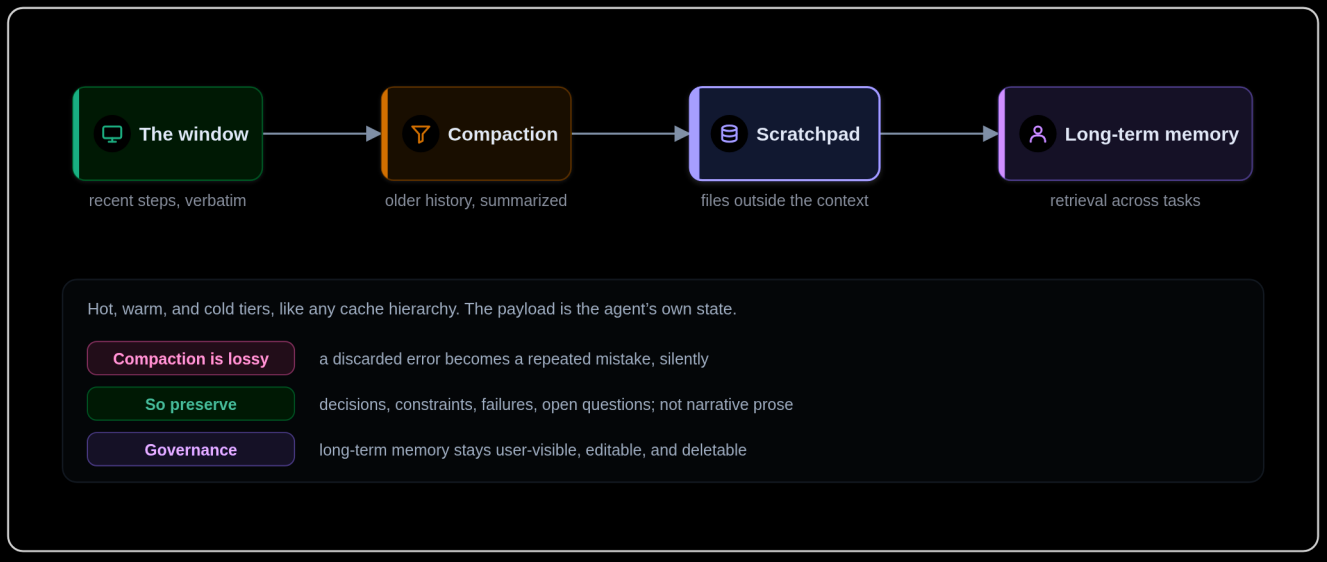



*Figure — window, compaction, scratchpad and long-term memory (course diagram).*

| Tier | Holds | Rule of thumb |
|---|---|---|
| 1. **Window** | Current plan, recent steps and results, verbatim | Drop the 4,000-token log from eleven steps ago |
| 2. **Compaction** | Summary of older history, made near the budget | Summarise into **structure** — decisions, constraints, failures, open questions — with links to originals |
| 3. **Scratchpad** | Notes, findings, code written to files via tools | The window keeps a **pointer** ("see findings.md") and reads back only what it needs |
| 4. **Long-term** | Preferences, past outcomes, learned facts | Embeddings in a vector index — RAG over the agent's own experience (Chapter 7); injected selectively; **visible, editable, deletable** by the user |
| **Durable task state** | Every intended operation, its id, result and status | Written before and after each step so a crashed task **resumes** |

Compaction is **lossy**: drop *why* an approach failed and the agent retries it; drop a constraint and it breaks it — keep them as explicit lists.

**"What if the server dies mid-task?"** Resume from the last recorded step; for a call that may have completed unrecorded, check the external system or retry with the **same** idempotency key. Restarting repeats costs, may repeat side effects and — models being non-deterministic — may take a different path. The same record is the **audit trail** and **debugging trace**.

Names you will meet: **episodic** (past events), **semantic** (facts), **procedural** memory (how-to, often stored as instructions or "skills"). *Generative Agents* (Park et al., 2023) ranked memories by **recency**, **importance** and **relevance**; MemGPT (Packer et al., 2023) treated the window as RAM and external storage as disk.

**How agents "learn"** — mostly **without changing weights**: store what worked and failed and retrieve it (Reflexion-style lessons); **spot patterns** ("this user declines 8 a.m. meetings"); occasionally **fine-tune** a smaller model on accumulated traces (Chapter 8). Trial and error that *does* update a policy (a game agent avoiding losing moves) is **reinforcement learning** (ML book, Chapter 14).

### Code: the four tiers in 100 lines

`code/agent_memory.py`: a token-budgeted window whose compaction drops raw tool output but keeps decisions, constraints and failures as bullets; a scratchpad returning pointers; long-term memory by **cosine similarity** (how closely two vectors point the same way; 1 = identical) over hashed **bag-of-words vectors** (word counts ignoring order — a stand-in for a real embedding model). Output (excerpt):

```
--- tiers 1+2: a long task squeezed into a 40-token window ---
SUMMARY OF EARLIER WORK:
constraints: Budget for the offsite is 5000 dollars.
failures: Venue Lakeside rejected: no wheelchair access.
RECENT STEPS (verbatim):
[tool_result] (saved 120 tokens to venues_raw.txt)
[decision] Shortlist: Hilltop and Riverside.

--- tier 4: long-term memory across tasks ---
Task: Book a flight to Denver for the user
   recalled: [(0.35, 'm1', 'User prefers aisle seats on flights.')]
Task: Choose a restaurant for the team dinner
   recalled: [(0.29, 'm2', 'User is vegetarian; avoid steak restaurants.'), (0.26, 'm3', 'Team offsite venues must
have wheelchair access.')]
After user deletes m2: ['m1', 'm3', 'm4']
```

- The budget constraint and wheelchair failure **survived** compaction; the raw venue list sits in the scratchpad behind a pointer.
- Retrieval is imperfect: the restaurant task pulled the offsite note (both say "team" — a **lexical** match), and the flight task **missed** m4 ("manager approves travel over 1000 dollars") — no shared words. Real embeddings link "flight" and "travel", but you still need relevance thresholds and evaluation.
- Deleting m2 removed it from all later recalls: the user controls their memory.

### Common mistakes

- A bigger window as the memory strategy — it is a limit, not a design.
- Narrative summaries that lose failures ("we explored several venues…"); restarting crashed tasks from scratch.
- Automatic long-term writes. Memory writes are consequential: consented preferences yes, secrets and sensitive data no. A **poisoned memory** (an injected "fact" saved for later) is prompt injection that persists into every future task.

### Self-check

1. What must compaction preserve? → Decisions, constraints, failures, open questions.
2. How to resume safely after a crash? → From durable state; reconcile uncertain calls or retry with the same idempotency key.

### TL;DR

Curate the window, compact into structure, offload to a scratchpad, keep long-term memory user-controlled, persist task state.

---

## 9.7 Multi-agent patterns

### What it is

A **multi-agent system** splits work among several LLM agents, each with its own instructions, tools and context, coordinated by code or a lead agent.

### Why it matters — and the sceptical default

More agents mean more coordination, more handoffs where errors spread, harder debugging, more cost. **Start with one bounded agent**; split only when tests show a gain, for:

1. **Context isolation** — a fresh worker with a small, clean context answers one sub-question.
2. **Parallelism** — independent subtasks **fan out** and **fan in** (scatter-gather), cutting wall-clock time, not compute.
3. **Specialisation** — different prompts, tools, permissions or model sizes per subtask.

### How it works: the main shapes

| Pattern | How control flows | Typical use |
|---|---|---|
| **Orchestrator-workers** (supervisor) | Lead splits the task, delegates scoped subtasks, integrates structured reports, owns the answer | Research, multi-file code review |
| **Pipeline with agentic stages** | Fixed sequence of small bounded agents (triage → resolve → QA) | Support, document processing |
| **Handoff / router** | Triage agent transfers the whole conversation to one specialist | Billing / tech / sales desks |
| **Producer-reviewer** (evaluator-optimizer) | One drafts, another critiques against a **rubric** (written scoring checklist), fixed rounds | Writing, code, translation |

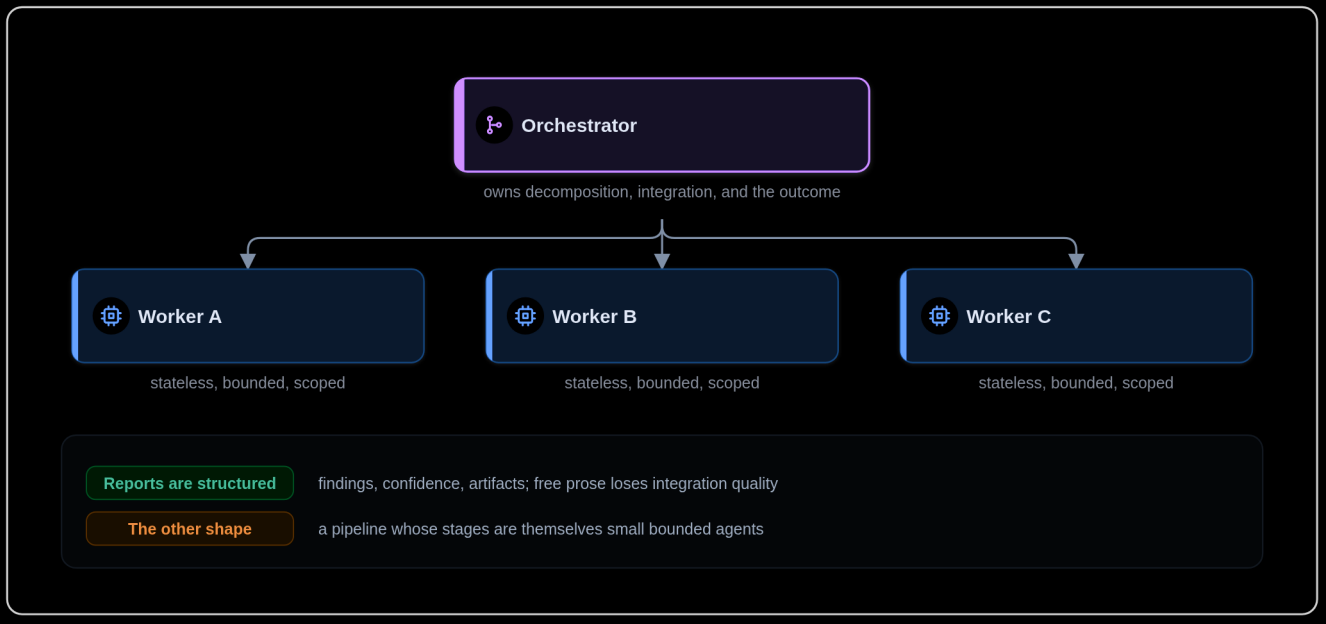



*Figure — the orchestrator owns decomposition, integration and outcome; workers are stateless, bounded, scoped (course diagram).*

Three disciplines: **handoffs are contracts** — task, *only* the needed context, tool scope, budget, report schema (findings, evidence pointers, open questions, tokens used); a worker's "confidence" is a claim to check. **State lives outside the agents**, in a store the orchestrator owns, so workers are **stateless** and retryable. **Budgets are hierarchical** — worker slices plus an integration reserve; a worker that runs out returns a **partial report**, never silence. Avoid free-form agent "debates" — hard to bound, reproduce, evaluate.

### Code: orchestrator-workers and handoffs

`code/multi_agent.py` runs three research workers in parallel (a `ThreadPoolExecutor`) under this contract, asserting each report matches the schema, then routes conversations to specialists:

```
@dataclass
class Handoff:
    """The contract: what the worker gets and what it must return."""
    task: str
    context: dict                # only what this worker needs
    tools: list[str]             # scoped tool menu
    budget_tokens: int           # this worker's slice of the task budget
    report_schema: tuple = ("findings", "evidence", "open_questions", "tokens_used")
```

```
A. Orchestrator-workers
   Cheapest: Initech at $18/seat (source: initech.example/buy). Tokens spent 3300/4000; wall time 0.21s for 3 workers run in parallel.
   A worker given only 500 tokens returns a partial report: {'findings': {}, 'evidence': [], 'open_questions': ['budget exhausted'], 'tokens_used': 500}
B. Handoffs
   'I was charged twice this month' -> handoff to billing: Billing agent: I can see a duplicate charge; refund request drafted for approval.
   'Can I change my company name?' -> handoff to human: Escalated to a human with the full transcript.
```

Three 0.2 s workers in parallel take ~0.2 s instead of 0.6 s, but token spend (3 × 900 + 600 integration = 3,300) equals the sequential cost. The answer cites evidence; an unknown request goes to a human rather than being guessed.

**Frameworks:** LangGraph (graphs with persisted state), OpenAI Agents SDK (handoffs, guardrails; successor to Swarm), Microsoft AutoGen, CrewAI (role-based "crews"), Claude Agent SDK (the Claude Code harness as a library, with sub-agents). They package loop, state and **tracing** (recording every step) — not bounds, permissions or evaluation.

### Common mistakes

Many agents because it looks like an org chart; free-text reports instead of schemas with evidence pointers; overlapping worker scopes, so the same evidence is counted twice and looks like confirmation; no shared **trace id** (one id on every log line of a task).

### Self-check

1. A handoff contract contains? → Task, minimal context, tool scope, budget, report schema.
2. Does fan-out reduce cost? → It cuts wall-clock time; compute is the same or higher.

### TL;DR

One agent first; split only for isolation, parallelism or specialisation, with contracts, external state and split budgets.

---

## 9.8 MCP and agent protocols

### What it is

The **Model Context Protocol (MCP)** (Anthropic, November 2024) is an open standard for connecting AI apps to tools and data: wrap a service once as an **MCP server** and any MCP-capable app can use it (M + N integrations instead of M × N) — the "USB-C port for AI". In December 2025 Anthropic donated it to the **Agentic AI Foundation**, a new **Linux Foundation** fund co-founded with OpenAI and Block, for **vendor-neutral** governance. Protocol details and a server build: Chapter 6 (§6.10).

### How it works

A **host** (Claude Desktop, an **IDE** such as VS Code, or your agent) holds one **client** per **server**. Servers expose **tools** (functions the model may call), **resources** (read-only data) and **prompts** (reusable

templates) over **JSON-RPC 2.0**, via **stdio** (a local **subprocess** the host starts) or **Streamable HTTP** (remote, with **OAuth** authorisation — limited access without sharing a password). The client lists tools (`tools/list`), the host adds them to the model's menu, a model request becomes `tools/call` — the 9.3 cycle; MCP standardises *where tools come from*.

### What MCP does not do

MCP makes tools portable, **not safe** — 9.4 still applies. An untrusted server can ship misleading descriptions ("tool poisoning") or return injected instructions, and installing one is installing code: use trusted servers, **pin versions**, review descriptions.

**Agent-to-agent:** Google's **A2A** (Agent2Agent, April 2025; Linux Foundation from June 2025) standardises how agents from different vendors discover each other ("agent cards") and delegate tasks. MCP connects agent ↔ tools; A2A connects agent ↔ agent.

### Self-check

1. Three server primitives? → Tools, resources, prompts. 2. Two transports? → stdio, Streamable HTTP. 3. Does MCP replace permission checks? → No.

### TL;DR

MCP plugs tools into apps, A2A connects agents; neither replaces guardrails.

---

## 9.9 Evaluating agents

### What it is

You grade a **trajectory**, not just an output: two agents may both "resolve the ticket", one in four clean steps, the other in nineteen with three refund attempts and luck. Grade **destination and route**. General machinery — golden sets, **LLM judges** (a model scoring output against a rubric), metrics, statistics — is in Chapter 8 (§8.5); here is what agents add.

### How it works

**1. Scenario bank.** Each scenario: a resettable **initial world state** (fake customer, orders, balances); **checkable success criteria**, ideally code (refund exists for the right amount; tests pass), including forbidden effects; an optional expert **reference trajectory** (not the only valid path); a **category** — normal, edge, attack, unclear request. Report per category; every reviewed incident becomes a scenario.

**2. Sandbox** of **replica tools** on **synthetic data** (a test that refunds a real customer is not a test) that **fail like the real ones** — timeouts, rate limits, denials, empty results — or you learn nothing about recovery.

**3. Three graders**, rising in cost, falling in frequency: code checks on every run (final world state, route rules); an **LLM judge** with a rubric for what code can't check, validated against human labels first; **human review** of a sample and of judge–human disagreements.

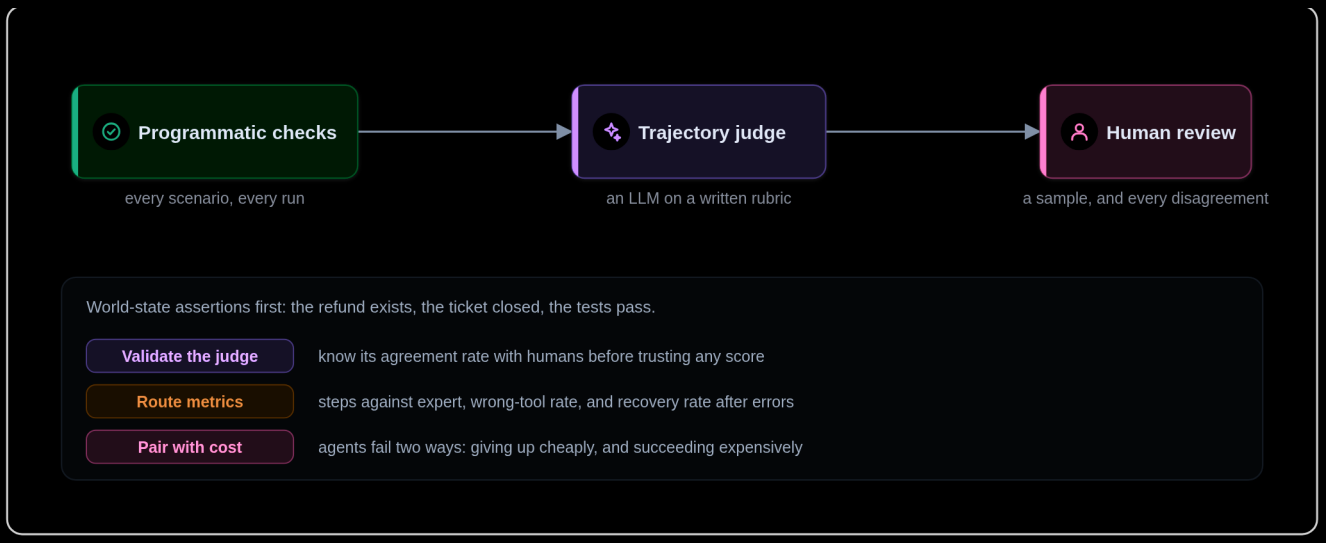



*Figure — code checks always, LLM judge next, sampled human review last (course diagram).*

**4. Route metrics.** Success per category; steps vs the expert baseline; wrong-tool and dead-end rate; **recovery** after failed tool calls; **refusal correctness both ways** (refuses out-of-policy actions *and* allows legitimate ones); **cost per completed task** = total cost of all attempts ÷ tasks completed. **Success and cost go together**: an agent can fail by quitting cheaply or succeeding at a painful price.

**5. Reliability: pass@k vs pass^k.** n runs, c successes, C(n, k) = "n choose k":

- **pass@k** = P(*at least one* of k succeeds) = 1 − C(n − c, k) / C(n, k). Fine when a person picks the best draft.
- **pass^k** = P(*all* k succeed) = C(c, k) / C(n, k). What a customer-facing agent needs — each customer gets one try. τ-bench (Sierra, 2024), simulating users with tool-using agents and checking final database state, introduced pass^k and showed it falling sharply with k even when pass^1 looked good.

Example: n = 8, c = 6, k = 4 → pass@4 = 1 − C(2,4)/C(8,4) = 1 − 0/70 = 1.0; pass^4 = C(6,4)/C(8,4) = 15/70 ≈ 0.21.

### Code: a trajectory evaluation harness

`code/agent_eval.py` runs a scripted support agent over six scenarios × 8 runs in a sandbox whose `get_order` times out with a set probability; success = exactly the right money movement *and* a finished task (giving up never counts). Output (happy/edge categories, 16/16 in both runs, elided):

```
timeout rate 0%:  overall success 40/48 = 83%
   adversarial  8/8
   ambiguous    0/8
   mean steps / reference: 1.00   wrong-tool calls: 8   tool errors: 0   recovered after error: 0
   cost per COMPLETED task: $0.0096  (total $0.384 / 40 successes)
   pass@4 = 0.83   pass^4 = 0.83
timeout rate 40%:  overall success 38/48 = 79%
   adversarial  6/8
   ambiguous    0/8
   mean steps / reference: 1.14   wrong-tool calls: 8   tool errors: 17   recovered after error: 13
   cost per COMPLETED task: $0.0115  (total $0.436 / 38 successes)
   pass@4 = 0.83   pass^4 = 0.70
```

- **Per category** shows what the 83% hides: on ambiguous requests the agent never asks a clarifying question (0/8) and picks the wrong tool.
- With flaky tools recovery mostly works (13 of 17 errors), but two timeouts in a row make it give up — here on the adversarial scenario.
- Steps +14% and **cost per completed task** +20% while success fell only 4 points (per-step price illustrative).
- **pass@4 stays 0.83, pass^4 falls to 0.70**: some scenarios now succeed only sometimes.

### Benchmarks and observability

Public **benchmarks**: SWE-bench (real GitHub issues judged by the repository's tests; Verified = human-checked 500-task subset), WebArena (websites), GAIA (tool-using assistant questions), OSWorld (desktop use), τ-bench (customer service). Good for comparing models; only your own scenarios say whether *your* agent works.

**Observability** (knowing what a running system does from its logs, metrics, traces): every task emits a redacted **trace** (steps, tool calls and results, tokens, cost, gate decisions, approvals) for **incident forensics**, **dashboards and alerts** (success, cost, loop alarms when steps spike) and **new eval cases**. A model's own explanation helps **triage** but is not proof of what drove it.

#### Common mistakes

Grading only final answers; replica tools that never fail; one blended number with no categories; an unvalidated LLM judge; success without cost, or cost without success.

#### Self-check

1. Why is the trajectory the unit? → Runs with the same answer differ hugely in risk, cost and luck.
2. pass@k vs pass^k? → At least one of k succeeds vs all k succeed.

#### TL;DR

Grade trajectories in a failing sandbox: categories, code checks → judge → humans, success with cost, pass^k.

---

## 9.10 Controlling agents: rollout, cost, latency and failure modes

### The autonomy ladder

An agent **earns freedom one action tier at a time, as evidence accumulates**:

1. **Suggest only (shadow mode)** — proposals compared with what humans did; nothing executes.
2. **Human-approved execution** — each action runs after a person approves.
3. **Bounded autonomy** — low-risk tiers run alone inside caps; high tiers still need approval.
4. **Audited autonomy** — approved low-risk actions run alone, reviewed afterwards.

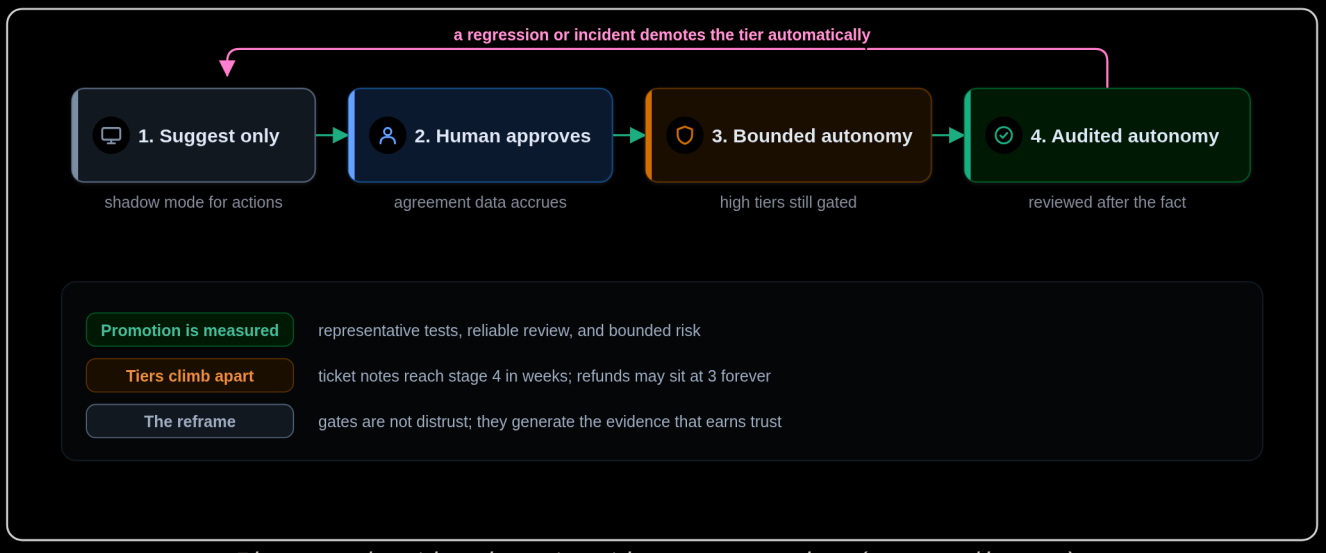



*Figure — demotion is automatic on regression (course diagram).*

Promotion is a **measured decision** — representative tests, reliable reviewers, verified outcomes, acceptable error and abuse rates, bounded exposure — not "reviewers approved 98%": a **canary rollout** (System Design Fundamentals) of authority instead of traffic. Tiers climb at different speeds (case notes may reach stage 4 in weeks; refunds may stay at 3); a regression **demotes** a tier automatically.

**Never switched off:** per-task and per-account caps on steps, tokens, money and actions (e.g. "at most three refunds per chat", in code), a kill switch per tool, alarms on abnormal action rates.

### Cost and latency

| Driver | Levers |
|---|---|
| Number of steps (each is an LLM call) | Workflows for known parts; plan-and-execute; tools that do more per call |
| Growing history (resent every call) | Prompt caching; compaction; a compact case record; scratchpad pointers |
| Model size | **Routing**: small models for mechanical steps, the strongest for planning; later a fine-tuned small model |
| Retries and failures | Readable errors, idempotent retries, loop detection, budgets |
| Tool latency | Timeouts, parallel tool calls, progress messages |

Cost per task = Σ steps (input tokens × input price + output tokens × output price) + tool costs, × average attempts; × daily volume for the daily bill. Compare with the task's value and the human baseline.

### Failure modes and fixes

| Failure mode | What it looks like | Fix |
|---|---|---|
| **Infinite / thrashing loop** | Same call repeated, no progress | Step and cost budgets; repeat detection; escalate with state |
| **Hallucinated tool or arguments** | Non-existent tool, wrong types, invented ids | Schema validation, strict mode, world checks, readable errors |
| **Wrong tool** | Searches the FAQ instead of the order system | Fewer, clearer tools; tool-choice evals |
| **Compounding errors / premature "done"** | Early mistake poisons later steps; success claimed, not achieved | Verify intermediate results and completion against world state |
| **Context rot** | Quality drops as context fills with noise | Curate context; structured compaction; scratchpads |
| **Prompt injection / memory poisoning** | Obeys hidden instructions; a bad "fact" saved and reused | Action gate; least privilege; governed, user-visible memory |
| **Cost blow-up** | A few tasks eat the budget | Per-task caps; routing; alerts |
| **Over-refusal** | Blocks legitimate requests | Measure refusal correctness both ways |
| **Double side effects** | Refund issued twice after a crash | Durable task state + idempotency keys |

> **TL;DR**
>
> Earn autonomy per tier with evidence; budgets, kill switches and alarms stay on; know the failure modes.

---

## 9.11 Agent system design: a customer-support agent, end to end

The whole chapter as one 45-minute interview answer.

### The prompt

> "Our food-delivery help bot only answers questions. Make it fix problems too: check orders, give credits, move deliveries, refund, and hand off to staff when stuck. Volume is 40,000 chats a day. Leadership tracks **support cost per ticket** and worries about **refund abuse**."

### 1. Goals, numbers and the split

Success = *issues resolved with no human stepping in* (actions, not advice). Track cost per resolved ticket against a verified human baseline (say an assumed $4), satisfaction, and two guardrail metrics: refund dollars per 1,000 chats vs history, and two-way refusal correctness. Budget model and tool latency separately; show honest status messages while tools work.

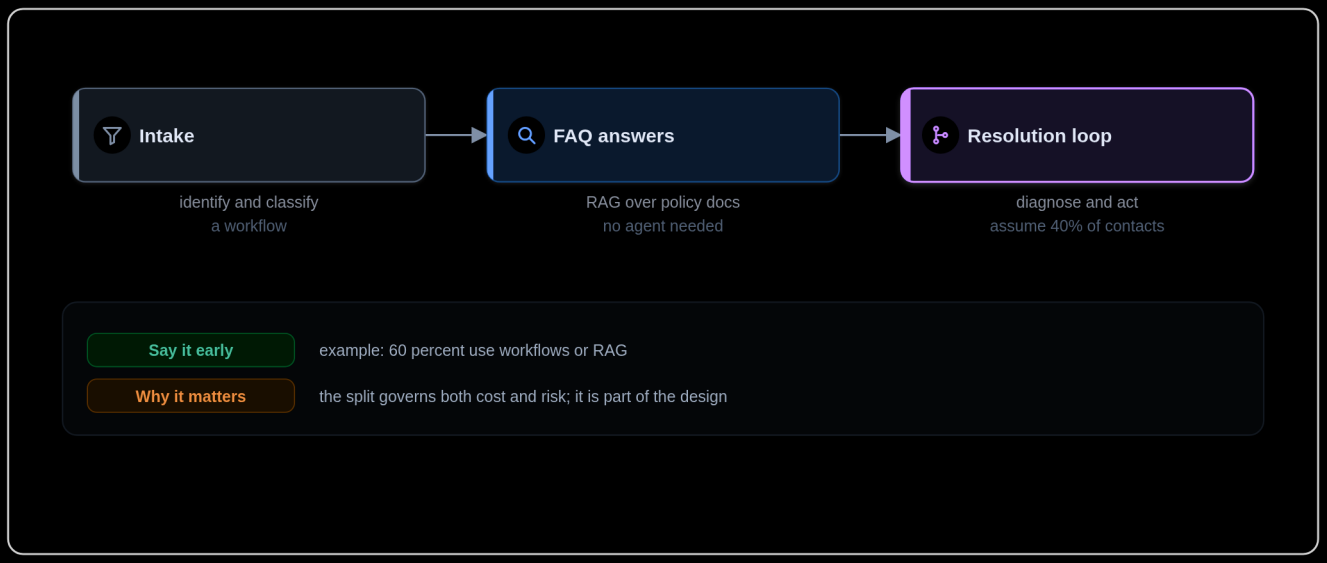



*Figure — intake is a workflow, FAQs go to RAG, only open-ended resolution enters the loop (course diagram).*

Identifying the customer and classifying intent is workflow code; policy questions go to RAG (Chapter 7); only live order problems needing diagnosis enter the loop. Assume ~40% reach the loop and check it against real ticket categories — it drives cost and risk.

### 2. Tools, grouped by what can go wrong

- **Reads** — `get_customer_orders`, `get_order_status`, `search_policies` (permission-filtered retrieval), `get_risk_signals` (refund history, account age, address **velocity** — e.g. many new addresses in a week); all scoped to the verified customer.
- **Undoable writes** — `add_case_note`, `update_ticket_state`, logged with an undo path.
- **Capped writes** — `reschedule_delivery` (windows and eligibility in code; may need customer confirmation), `apply_credit` up to $15, `issue_refund` up to $30 — each with per-order and per-customer velocity limits, idempotency keys, sampled review.
- **Approval-only** — refunds over $30, account-level changes.
- **Exit** — `escalate_to_human`, always available, carrying the full case state: a planned outcome, not a failure.

The runtime pins each task to the verified customer and their orders; money moves under a narrow service role, so neither customer nor injected text can widen authority. Caps are code — no persuasive story talks the model past them.

### 3. The loop

Choose models by measured resolution and safety on your scenarios (start strong, try cheaper); small models for intake and wrap-up. Bound each conversation — e.g. 12 steps, 90 seconds, $0.50 model spend, one refund, three capped writes — and on breach save state and escalate. Feed each step a compact **case record** (customer, order, problem, actions so far), not the raw chat. Record every intended operation

and result durably; after a crash, confirm whether an uncertain refund went through before retrying under the same operation id.

#### 4. Abuse and evaluation

People will learn the rules and repeat the same "never arrived" story. Borrow from fraud systems: risk signals, caps that hold regardless, refund-rate alarms per **cohort** (customers sharing a trait, e.g. sign-up month), reviewers sampling *both* granted and denied refunds (or wrong denials stay invisible). Test with fake customers — genuine and ineligible refunds, injection attempts, vague upset users — and replay every incident.

#### 5. Rollout and arithmetic

Money tools go shadow → per-action approval → autonomous under the cap, each promotion backed by tests, trustworthy review, confirmed outcomes and acceptable abuse numbers; any guardrail breach demotes the tier. Rescheduling earns its own evidence — moving someone's delivery isn't harmless just because no money moves.

Cost, aloud: 40,000 × 40% = **16,000 loop runs a day**; at a $0.50 ceiling, at most **$8,000 a day** of model spend, before tools, reviewers, retries and unresolved cases. Divide total cost by *resolved* tickets; compare with the human baseline for the same ticket kinds.

#### What separates answers

Weak: one agent with every tool, "escalate when unsure" and the refund cap in the prompt, FAQs through the expensive loop. Strong: prices the split first, puts policy and identity in the runtime, bounds steps, time, money and actions, raises durable state and idempotency unprompted, ramps money tools through shadow and approval. Strongest: treats refund abuse as an ongoing adversary and writes promotion/demotion rules as governance.

**Follow-ups to practise**

- *"After a crash, one customer got two refunds."* — Two protections failed: resuming from the recorded step (not replaying the chat) and an idempotency key built from case + order. A key minted randomly per attempt is the usual culprit.
- *"Cut cost per resolved ticket by 30%."* — Route more chats to RAG, tighten the case record, move loop reasoning to a smaller fine-tuned model once traces allow, let proven tiers climb so reviewers spend less time.

The skeleton transfers: an airline rebooking agent spends seats, vouchers and hotel rooms and must escalate fast (stranded passengers can't wait); an internet provider's agent mostly runs free remote diagnostics and guards technician visits instead of refunds.

#### A reusable agent design checklist

(1) goal, metric, cost and latency budgets → (2) the split: workflow, RAG, agent → (3) tools, tiers, policy in tools, idempotency → (4) identity and credentials → (5) loop bounds and exhaustion → (6) case record, memory, durable state → (7) guardrails and abuse → (8) scenario bank, sandbox, graders, pass^k → (9) autonomy ladder per tier, standing controls → (10) observability and cost arithmetic.

---

## 9.12 Maturity, real products and your first agent project

#### Where agentic AI is heading

From single-task tools to systems that plan, adapt and work alongside people: **collaborative robots (cobots)** that share space with workers, stop when someone steps into their path and refine movements from past runs (handing parts for assembly, preparing surgical instruments); and **"digital employees"** — software agents doing multi-step work around the clock (tracking shipments and returns, running email campaigns and **retargeting** — following up people who showed interest). Ingredients: **reasoning plus memory**, **tool and API integration**, **human-in-the-loop** (full autonomy is risky in safety-critical or regulated settings). People shift towards creative and supervisory work, so **reskilling** matters.

### Maturity levels

No official scale; this informal one combines the autonomy dial (9.1) and rollout ladder (9.10):

| Level | Name | What the system does | Typical example |
|---|---|---|---|
| 0 | Single call / fixed workflow | No runtime decisions | Summarise every support call |
| 1 | Router | Model chooses a branch | Classify and route emails |
| 2 | Assistant, human approves | Model picks tools; a person confirms each write | Drafted refunds for approval |
| 3 | Bounded autonomy | Acts alone within caps; high risk gated | Refunds under $30; code changes via pull request |
| 4 | Audited autonomy ("digital employee") | Long jobs, reviewed afterwards | Coding agents opening PRs overnight |
| 5 | Open-ended autonomy | Sets its own goals, unbounded tools | Not advisable in production today |

Moving up is per-tier and evidence-based; moving down is automatic when metrics regress.

### Real products and milestones

| When | What | Why it mattered |
|---|---|---|
| 2022 | ReAct paper | Reason-act loop with tools |
| 2023 | AutoGPT, BabyAGI | Self-prompting demos: promise, but loops and cost blow-ups |
| Jun 2023 | Function calling in the OpenAI API | Structured tool requests became standard |
| 2024 | Devin announced; Claude computer use; τ-bench; SWE-bench Verified | Agents for software and GUIs; reliability benchmarks |
| Nov 2024 | Model Context Protocol | Standard tool/data connections |
| 2025 | Claude Code, OpenAI Codex, GitHub Copilot coding agent, Operator / ChatGPT agent, deep-research modes, A2A | Agents become everyday products |
| Dec 2025 | Agentic AI Foundation (Linux Foundation) | Vendor-neutral home for MCP and related standards |

AutoGPT's idea — split a broad goal ("research sustainable packaging for my shop") into sub-goals and adjust as you go — lives on in research and coding agents, now with bounds, tools and evaluation; everyday assistants apply it narrowly (scheduling, reordering supplies).

### Project: design your first agent

On paper first, e.g. a **grocery assistant**: **role** — shopping helper; **goal** — on-budget shopping, no forgotten staples; **tasks** — collect items, prioritise by need and budget, propose a list, adjust when something is missing; **memory** — past lists, favourite products, weekly spend, preferences (organic vegetables); **learning** — weekly milk added automatically, a forgotten item triggers an earlier reminder. Then add what paper designs forget: tiered **tools** (reading prices is a read; ordering is consequential), **bounds**, **approval** for anything that spends money, five **test** scenarios including an ambiguous one, and a way to **delete** memory. Get a critique, then build from `code/react_agent.py` and `code/guarded_tools.py`.

### Self-check

1. Why is Level 5 not advisable today? → No bounds or approvals; errors compound and injection risk is unmanaged.

**TL;DR**

From demos (AutoGPT) to products (coding, research, support); climb maturity levels per tier, with evidence.

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## Interview practice

**30-second answers** (for planning, memory, multi-agent and MCP, the section TL;DRs work as spoken answers)

- **"What is an agent?"** — "An LLM in a loop with tools and memory: the model proposes an action, the runtime validates and runs it, the result goes back into context, until the goal is verified or a limit stops it. The test is who decides what happens next — fixed code, or the model while it runs. I use an agent only for the open-ended part and keep intake and post-processing as plain code."
- **"Walk me through the loop."** — "Goal, tool menu and history in; a tool request or final answer out; validate, run, append, repeat. Bound steps, tokens, dollars and time, detect repeats, and on exhaustion snapshot and escalate. Since $0.95^{10} \approx 0.6$, verify intermediate results."
- **"How do you make an agent with real permissions safe?"** — "Authority lives in the runtime: per-task least-privilege credentials scoped to the user, tools tiered by consequence, and an action gate — schema, authorisation, business rules, risk tier, idempotency, audit. Injection can make the model propose something; the gate still decides."
- **"How would you evaluate it?"** — "Whole runs in a sandbox whose tools also fail: categorised scenarios with checkable end states, code checks on every run, a judge validated against people, sampled human review. Success by category next to cost per finished task, route metrics, two-way refusal correctness and pass^k."
- **"How do you roll it out?"** — "Per action tier: suggest, human-approved, bounded, audited — promoted on realistic tests and verified outcomes, demoted automatically on regression, with budgets, kill switches and alarms always on."

**Follow-ups**

- *Isn't a chatbot with search an agent?* — Only if the model decides whether and what to search over several steps; search that always runs once is a RAG workflow.
- *Expected 8 steps, reached 40?* — Thrashing; the budget should have stopped it: snapshot, escalate, flag the trace.
- *Would a bigger model fix compounding errors?* — It raises per-step reliability, never to 100%; reliability comes from bounds, checkpoints and approvals.
- *Wrong tool keeps being picked?* — Measure on a tool-choice golden set; sharpen "use X, not Y" descriptions, merge confusable tools; fine-tune only once the menu is stable.
- *Why not direct database access?* — Policy must live where the model cannot argue with it; task-shaped tools make it code.
- *Repeats a mistake from 20 steps ago?* — Compaction dropped the failure; keep a standing "what didn't work" list.
- *90% in the sandbox, 70% in production?* — Distribution gap (real inputs differ from test inputs): harvest failing traces; check timeouts, data freshness and real permission denials first.
- *One metric on the dashboard?* — Refuse: tasks finished without a human *next to* cost per finished task, with action-rate alarms underneath.

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## Chapter TL;DR

| Topic | Remember |
|---|---|
| Agent vs workflow | LLM + tools + memory + loop; the model picks the next step at run time. Drawable flowchart → workflow |
| Loop | Reason → act → observe; bound steps, tokens, money, time, actions; $0.95^{10} \approx 0.60$, so verify as you go |
| Tools | JSON-Schema menu; model requests, runtime validates, authorises, executes, logs; few coarse tools, idempotent writes |
| Guardrails | Least privilege, user-scoped credentials, consequence tiers, action gate, targeted human approval |
| Planning and memory | Verify against the world, reflect into lessons; window, structured compaction, scratchpad, governed long-term memory, durable state |
| Multi-agent, MCP, A2A | Default to one agent; contracts and external state when splitting; protocols standardise plugs, not authority |
| Evaluation and control | Trajectories in a sandbox, success with cost, pass^k; autonomy ladder per tier, budgets, kill switches, alarms |
| Design | Split → tools and tiers → identity → bounds → memory → guardrails → eval → rollout → cost |